

NETWORKING

Complete Coursebook

Terminale / Year 13 - 80 pages - Explanations in English

Core coverage

- Network foundations, OSI/TCP-IP and protocols
- Ethernet switching, VLANs, IPv4/IPv6 and subnetting
- Routing, DHCP, DNS, wireless, VPNs and security
- Cloud networking, troubleshooting, labs and exam preparation

Purpose

This coursebook introduces the principles and practical skills needed to understand, design and troubleshoot modern computer networks. It is suitable for upper-secondary, pre-university and introductory technical study.

Learning approach

Each lesson combines a focused explanation, key concepts, a short reference table, a worked or applied example, and practice questions. Revision pages at the end consolidate the most important skills.

Recommended preparation

Learners should be comfortable with basic computer use and simple binary arithmetic. Programming knowledge is helpful but not required for most networking topics.

Course outcomes

- Explain how layered network protocols cooperate.
- Calculate IPv4 subnets and interpret IPv6 addressing.
- Describe switching, routing and key network services.
- Apply basic security, wireless and troubleshooting principles.
- Read diagrams, packet information and simple device configurations.

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A practical method

- Read the concept first, then redraw it as a small diagram from memory.
- For subnetting, always show the prefix, block size, network, host range and broadcast address.
- Use packet captures and command outputs as evidence, not guesses.
- Troubleshoot from the physical layer upward unless symptoms point clearly elsewhere.
- Keep a vocabulary list of protocols, ports, acronyms and command terms.
- Review security implications whenever you design connectivity.

Core commands to recognize

```
ping 192.168.1.1
tracert example.com      # tracert on Windows
ip addr / ipconfig
ip route / route print
nslookup example.com
arp -a
netstat -an
ssh admin@router
```

Important note

Command syntax varies by operating system and network platform. The goal of this course is to understand what information a command reveals and how that evidence supports troubleshooting decisions.

Lesson 1: What Is Computer Networking?

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FOUNDATIONS

Core idea

Computer networking is the practice of connecting devices so that they can exchange data and share resources. Networks range from small home LANs to global carrier and cloud infrastructures. Good network design balances performance, availability, security, cost and manageability.

Key points

- Nodes include computers, phones, servers, printers, routers and sensors.
- Links may be copper, fiber or wireless radio.
- Protocols define the rules used to exchange data.
- Network services include addressing, naming, authentication, storage and applications.

Key terms and examples

Term	Meaning	Example
Node	device participating in network	laptop
Link	communication medium	Ethernet cable
Protocol	communication rules	TCP/IP

Applied example

A school network may connect classrooms, servers, Wi-Fi access points and an internet router.

Practice

1. Explain the main purpose of what is computer networking?.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 2: LAN, WAN, MAN and PAN

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FOUNDATIONS

Core idea

Networks are often classified by geographic scope. A PAN serves an individual, a LAN serves a building or campus, a MAN spans a metropolitan area, and a WAN connects distant sites. The internet is a global internetwork made from many independent networks.

Key points

- LANs usually offer high speed and low latency.
- WAN links may use leased lines, MPLS, broadband, cellular or VPNs.
- The same IP protocols can operate across many physical media.
- Ownership and administration differ between private LANs and carrier networks.

Key terms and examples

Term	Meaning	Example
PAN	personal area	Bluetooth headset
LAN	local area	office network
WAN	wide area	branches in two countries

Practice

1. Explain the main purpose of lan, wan, man and pan.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
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- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 3: Network Topologies

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Core idea

A topology describes how devices and links are arranged. Physical topology shows actual cabling or radio relationships, while logical topology describes how data flows. Common patterns include star, bus, ring, mesh and hybrid designs.

Key points

- Star topologies are common in Ethernet LANs.
- Mesh designs improve resilience by providing multiple paths.
- A single point of failure can reduce availability.
- Topology affects cost, scalability and troubleshooting.

Key terms and examples

Term	Meaning	Example
Star	central connection point	switch-based LAN
Mesh	multiple paths	backbone network
Hybrid	combination	campus network

Practice

1. Explain the main purpose of network topologies.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 4: Network Performance Metrics

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FOUNDATIONS

Core idea

Network quality depends on several measurable factors. Bandwidth is capacity, throughput is achieved data rate, latency is delay, jitter is variation in delay, and packet loss is the proportion of data that does not arrive successfully.

Key points

- High bandwidth does not guarantee low latency.
- Real-time voice and video are sensitive to delay and jitter.
- Congestion can increase loss and latency.
- Good measurements should be taken under realistic traffic conditions.

Key terms and examples

Term	Meaning	Example
Bandwidth	maximum capacity	1 Gbit/s link
Latency	end-to-end delay	20 ms
Jitter	delay variation	3 ms

Practice

1. Explain the main purpose of network performance metrics.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 5: Network Devices Overview

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FOUNDATIONS

Core idea

Different devices perform different network roles. Switches connect devices within LANs, routers connect IP networks, access points provide wireless access, firewalls filter traffic, and modems or ONTs connect customer networks to service-provider media.

Key points

- Switches mainly use MAC addresses inside Ethernet LANs.
- Routers forward packets based on IP networks.
- Firewalls enforce security policy.
- Access points bridge wireless clients to a wired LAN.

Key terms and examples

Term	Meaning	Example
Switch	connects LAN devices	48-port switch
Router	connects networks	edge router
Firewall	filters traffic	perimeter firewall

Practice

1. Explain the main purpose of network devices overview.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 6: Client-Server and Peer-to-Peer

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FOUNDATIONS

Core idea

In client-server systems, dedicated servers provide services to clients. Peer-to-peer designs allow devices to share resources directly. Modern systems often combine both patterns, especially in distributed and cloud applications.

Key points

- Centralized services simplify management and policy.
- Peer-to-peer can reduce dependence on one central server.
- Authentication and authorization are easier to standardize in managed environments.
- Scalability and fault tolerance depend on architecture.

Key terms and examples

Term	Meaning	Example
Client	requests service	web browser
Server	provides service	web server
Peer	acts as client/server	file-sharing node

Practice

1. Explain the main purpose of client-server and peer-to-peer.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 7: Standards and Network Organizations

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FOUNDATIONS

Core idea

Networking works globally because devices follow shared standards. Important organizations include IEEE, IETF, ICANN, ISO and regional internet registries. Standards define Ethernet, Wi-Fi, internet protocols, identifiers and many operational practices.

Key points

- IEEE 802 standards cover LAN technologies such as Ethernet and Wi-Fi.
- IETF publishes internet standards as RFCs.
- ICANN coordinates domain names and global internet identifiers.
- Standards improve interoperability between vendors.

Key terms and examples

Term	Meaning	Example
IEEE	LAN/WLAN standards	802.3 Ethernet
IETF	internet standards	RFC documents
ICANN	names and identifiers	DNS coordination

Practice

1. Explain the main purpose of standards and network organizations.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 8: Network Documentation

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FOUNDATIONS

Core idea

Clear documentation is essential for operations and troubleshooting. Useful records include logical diagrams, physical diagrams, IP address plans, VLAN lists, cable labels, device inventories, configuration backups and change logs.

Key points

- Diagrams should distinguish physical and logical relationships.
- Consistent naming makes devices easier to identify.
- Configuration backups reduce recovery time.
- Change records help explain new faults after modifications.

Key terms and examples

Term	Meaning	Example
IP plan	address allocation	10.10.0.0/16
Inventory	device list	serial and location
Change log	modification history	router update

Practice

1. Explain the main purpose of network documentation.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 9: OSI Model Overview

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Core idea

The OSI reference model organizes communication into seven conceptual layers: Physical, Data Link, Network, Transport, Session, Presentation and Application. It is mainly a learning and troubleshooting framework rather than a literal implementation blueprint.

Key points

- Each layer provides services to the layer above.
- Encapsulation adds headers as data moves down the stack.
- Troubleshooting by layer helps isolate faults.
- Real protocols sometimes span more than one OSI layer.

Key terms and examples

Term	Meaning	Example
L1	Physical	signals and media
L2	Data Link	Ethernet frames
L3	Network	IP packets

Practice

1. Explain the main purpose of osi model overview.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 10: TCP/IP Model

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MODELS & PROTOCOLS

Core idea

The TCP/IP model describes the protocol architecture of the internet. A common four-layer view uses Link, Internet, Transport and Application layers. IP provides internetwork addressing while TCP and UDP provide transport services to applications.

Key points

- The Link layer handles local delivery.
- The Internet layer centers on IP and routing.
- The Transport layer provides end-to-end application delivery.
- Application protocols include HTTP, DNS, SMTP and SSH.

Key terms and examples

Term	Meaning	Example
Link	local network	Ethernet
Internet	internetwork delivery	IPv4/IPv6
Transport	process communication	TCP/UDP

Practice

1. Explain the main purpose of tcp/ip model.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 11: Encapsulation and Decapsulation

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MODELS & PROTOCOLS

Core idea

As application data is transmitted, each protocol layer adds control information. A TCP segment may be carried inside an IP packet, which is carried inside an Ethernet frame. The receiver removes these headers in reverse order.

Key points

- Headers contain addressing, control and integrity information.
- Encapsulation lets layers evolve independently.
- MTU limits can cause fragmentation or require path discovery.
- Packet captures expose encapsulated headers for troubleshooting.

Key terms and examples

Term	Meaning	Example
Segment	transport PDU	TCP segment
Packet	network PDU	IP packet
Frame	data-link PDU	Ethernet frame

Practice

1. Explain the main purpose of encapsulation and decapsulation.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 12: Ports and Sockets

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MODELS & PROTOCOLS

Core idea

Transport-layer ports identify application endpoints. A socket is typically described by protocol, IP address and port number. Servers listen on known ports, while clients usually use temporary ephemeral ports.

Key points

- Ports range from 0 to 65535.
- Well-known services commonly use standardized ports.
- A connection is identified by source and destination endpoint values.
- Firewalls often filter based on protocol and port.

Key terms and examples

Term	Meaning	Example
Port 80	HTTP	web
Port 443	HTTPS	secure web
Port 53	DNS	name resolution

Practice

1. Explain the main purpose of ports and sockets.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Core idea

TCP is a connection-oriented transport protocol that provides reliable, ordered byte delivery. It uses sequence numbers, acknowledgements, retransmission, flow control and congestion control to adapt to network conditions.

Key points

- The three-way handshake establishes a TCP connection.
- ACKs confirm received data.
- Retransmission handles loss.
- Windowing controls how much unacknowledged data can be sent.

Key terms and examples

Term	Meaning	Example
SYN	start connection	client request
ACK	acknowledge data	receipt confirmation
Window	flow-control value	receive capacity

Practice

1. Explain the main purpose of tcp fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 14: UDP Fundamentals

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MODELS & PROTOCOLS

Core idea

UDP provides connectionless datagrams with low protocol overhead. It does not guarantee delivery, ordering or retransmission. Applications choose UDP when speed, simplicity or real-time behavior is more important than built-in reliability.

Key points

- UDP adds source/destination ports and a checksum.
- Applications may implement their own reliability if needed.
- DNS commonly uses UDP for many queries.
- Voice/video applications can prefer timely delivery over retransmission.

Key terms and examples

Term	Meaning	Example
Datagram	independent message	UDP packet
Low overhead	minimal control	real-time traffic
No ordering	app responsibility	streaming design

Practice

1. Explain the main purpose of udp fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 15: ICMP and Diagnostic Messaging

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MODELS & PROTOCOLS

Core idea

ICMP carries control and diagnostic information for IP networks. Tools such as ping and traceroute rely on ICMP or related hop-limit behavior to test reachability and identify paths.

Key points

- ICMP echo request/reply supports reachability tests.
- Time Exceeded messages help traceroute discover hops.
- Destination Unreachable reports certain delivery failures.
- Blocking all ICMP can make troubleshooting and path MTU discovery harder.

Key terms and examples

Term	Meaning	Example
Echo	reachability	ping
Time Exceeded	hop limit expired	traceroute
Unreachable	delivery error	no route

Practice

1. Explain the main purpose of icmp and diagnostic messaging.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 16: Ethernet Fundamentals

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ETHERNET & SWITCHING

Core idea

Ethernet is the dominant wired LAN technology. It defines frame formats, MAC addressing and physical-layer options. Modern Ethernet typically uses switches and full-duplex point-to-point links rather than shared collision domains.

Key points

- MAC addresses identify Ethernet interfaces on a local segment.
- Ethernet frames include source and destination MAC addresses.
- Full duplex allows simultaneous send and receive.
- Different Ethernet standards support copper and fiber at many speeds.

Key terms and examples

Term	Meaning	Example
MAC address	Layer 2 identifier	00:11:22:33:44:55
Frame	Ethernet PDU	header + payload
Full duplex	two-way simultaneous	switch port

Practice

1. Explain the main purpose of ethernet fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 17: MAC Addresses and ARP

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ETHERNET & SWITCHING

Core idea

IPv4 hosts use ARP to discover the MAC address associated with a local IPv4 address. The result is stored temporarily in an ARP cache. IPv6 uses Neighbor Discovery instead of ARP.

Key points

- ARP operates only on the local broadcast domain.
- A host ARPs for the default gateway when the destination is remote.
- Stale or malicious ARP entries can cause connectivity problems.
- Dynamic ARP inspection can reduce spoofing risks in managed networks.

Key terms and examples

Term	Meaning	Example
ARP request	asks who owns IP	broadcast
ARP reply	returns MAC	unicast reply
ARP cache	temporary mapping	IP-to-MAC table

Practice

1. Explain the main purpose of mac addresses and arp.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 18: How Switches Learn

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ETHERNET & SWITCHING

Core idea

Ethernet switches learn source MAC addresses and associate them with incoming ports. They forward known unicast frames only to the correct port and flood unknown unicast, broadcast and some multicast traffic as needed.

Key points

- The MAC address table is learned dynamically.
- Entries age out after inactivity.
- Broadcast traffic remains inside a VLAN.
- Loops can cause frames to circulate indefinitely without protections.

Key terms and examples

Term	Meaning	Example
Learning	record source MAC	MAC table
Forwarding	send to known port	unicast
Flooding	send to multiple ports	unknown destination

Practice

1. Explain the main purpose of how switches learn.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 19: VLANs

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ETHERNET & SWITCHING

Core idea

A VLAN creates a separate Layer 2 broadcast domain on switching infrastructure. VLANs improve segmentation, reduce broadcast scope and allow logical grouping independent of physical location.

Key points

- Access ports normally carry one VLAN for end devices.
- Trunk links carry multiple VLANs between network devices.
- 802.1Q tags identify VLANs on trunks.
- Inter-VLAN communication requires Layer 3 routing.

Key terms and examples

Term	Meaning	Example
Access port	single VLAN	desktop port
Trunk	multiple VLANs	switch uplink
802.1Q	VLAN tagging	tagged frame

Practice

1. Explain the main purpose of vlans.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 20: Spanning Tree Protocol

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ETHERNET & SWITCHING

Core idea

Spanning Tree Protocol prevents Layer 2 loops by creating a loop-free logical topology. It elects a root bridge and places redundant links into forwarding or blocking roles so that backup paths can exist without causing broadcast storms.

Key points

- STP is essential where redundant Layer 2 paths exist.
- The root bridge influences forwarding topology.
- Rapid STP converges faster than classic STP.
- Incorrect edge-port settings can create outages or loops.

Key terms and examples

Term	Meaning	Example
Root bridge	reference switch	lowest elected ID
Blocked path	loop prevention	standby link
Convergence	topology stabilization	after link failure

Practice

1. Explain the main purpose of spanning tree protocol.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 21: Link Aggregation

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ETHERNET & SWITCHING

Core idea

Link aggregation combines multiple physical links into one logical connection. It can increase capacity and provide redundancy while simplifying forwarding decisions. LACP is a standard protocol used to negotiate aggregated links.

Key points

- All member links should use compatible settings.
- Traffic is distributed using hashing, not by splitting each frame.
- One flow may remain on one member link.
- Aggregation can protect against a single member-link failure.

Key terms and examples

Term	Meaning	Example
LAG	logical bundle	4 x 1G links
LACP	aggregation protocol	IEEE standard
Hashing	traffic distribution	source/destination tuple

Practice

1. Explain the main purpose of link aggregation.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 22: Power over Ethernet

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ETHERNET & SWITCHING

Core idea

Power over Ethernet delivers electrical power and data over compatible Ethernet cabling. It is widely used for access points, IP phones, cameras and IoT devices.

Key points

- PoE switches can simplify device installation.
- Power budgets must be checked across all powered devices.
- Different standards support different power levels.
- Cable quality and heat become important at higher power.

Key terms and examples

Term	Meaning	Example
PSE	power sourcing equipment	PoE switch
PD	powered device	IP camera
Power budget	available wattage	switch total

Practice

1. Explain the main purpose of power over ethernet.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 23: Ethernet Troubleshooting

Complete Networking Coursebook - explanations in English

ETHERNET & SWITCHING

Core idea

Layer 1 and Layer 2 faults are common causes of LAN outages. Troubleshooting should check link state, speed/duplex, cabling, VLAN membership, MAC learning, STP state and error counters.

Key points

- Start with physical indicators and interface status.
- Compare expected VLANs with actual port configuration.
- CRC errors can indicate cabling or signal problems.
- Packet captures help distinguish local from upstream faults.

Key terms and examples

Term	Meaning	Example
CRC error	frame integrity failure	bad cable
Duplex mismatch	incompatible link mode	poor throughput
Err-disabled	disabled port state	policy violation

Practice

1. Explain the main purpose of ethernet troubleshooting.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 24: IPv4 Addressing

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

IPv4 uses 32-bit addresses written as four decimal octets. An address contains a network portion and a host portion as defined by the subnet prefix length.

Key points

- CIDR notation writes prefix length after a slash.
- Private IPv4 ranges are used inside organizations.
- The network and broadcast addresses are not assigned to ordinary hosts in traditional subnets.
- A default gateway is a router address used for remote destinations.

Key terms and examples

Term	Meaning	Example
IPv4	32-bit address	192.168.1.10
/24	24-bit prefix	255.255.255.0
Gateway	next-hop router	192.168.1.1

Practice

1. Explain the main purpose of ipv4 addressing.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 25: Binary for Networking

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

Subnetting relies on binary arithmetic. Each IPv4 octet represents eight bits with place values 128, 64, 32, 16, 8, 4, 2 and 1. Converting between decimal and binary helps explain masks and address ranges.

Key points

- An octet ranges from 0 to 255.
- A subnet mask contains contiguous network bits in standard CIDR use.
- Bitwise AND identifies the network address.
- Powers of two determine subnet sizes.

Key terms and examples

Term	Meaning	Example
128	2^7	10000000
255	all eight bits set	11111111
AND	network calculation	IP AND mask

Applied example

Example: $192 = 128 + 64 = 11000000$ in binary.

Practice

1. Explain the main purpose of binary for networking.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 26: Subnet Masks and Prefix Lengths

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

A prefix length tells how many IPv4 bits belong to the network. For example, /24 leaves 8 host bits, while /26 leaves 6 host bits. Subnet size is $2^{(\text{host bits})}$, with traditional usable-host counts subtracting network and broadcast addresses.

Key points

- /24 corresponds to 255.255.255.0.
- /26 corresponds to 255.255.255.192.
- Longer prefixes create smaller subnets.
- Subnet size and host requirements should be planned before deployment.

Key terms and examples

Term	Meaning	Example
/24	256 addresses	254 traditional usable
/26	64 addresses	62 traditional usable
/30	4 addresses	2 traditional usable

Applied example

For /27, host bits = 5, so each subnet contains 32 addresses.

Practice

1. Explain the main purpose of subnet masks and prefix lengths.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 27: Subnetting Step by Step

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

Subnetting divides an address block into smaller networks. A reliable method is to determine host requirements, select a prefix, calculate block size, list network boundaries, then identify usable host ranges and broadcast addresses.

Key points

- Choose the smallest subnet that meets host needs plus growth.
- Block size reveals subnet boundaries.
- Verify that subnets do not overlap.
- Document network, gateway, host range and broadcast values.

Key terms and examples

Term	Meaning	Example
Need 50 hosts	choose /26	62 usable
Block size	64 addresses	0,64,128,192
Broadcast	last address	network + block - 1

Applied example

192.168.10.64/26 has usable hosts 192.168.10.65-126 and broadcast 192.168.10.127.

Practice

1. Explain the main purpose of subnetting step by step.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 28: VLSM

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

Variable Length Subnet Masking allows different subnet sizes inside the same address block. It conserves address space by allocating large subnets only where needed.

Key points

- Sort requirements from largest to smallest.
- Allocate each subnet on a valid boundary.
- Keep an address plan to prevent overlap.
- Routing protocols must support classless prefixes.

Key terms and examples

Term	Meaning	Example
VLSM	different prefix sizes	/25 + /27 + /30
Largest first	allocation strategy	200 hosts before 10
Classless	prefix-aware routing	CIDR

Practice

1. Explain the main purpose of vlsM.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Core idea

IPv6 uses 128-bit addresses written in hexadecimal. It greatly expands address space and changes several network behaviors, including neighbor discovery and address configuration.

Key points

- Leading zeros in each hextet may be omitted.
- One continuous run of zero hextets may be compressed with ::.
- IPv6 has no broadcast; multicast is used instead.
- Link-local addresses support communication on the local link.

Key terms and examples

Term	Meaning	Example
IPv6	128-bit address	2001:db8::10
Link-local	local segment	fe80::/10
Global unicast	internet-routable	2000::/3

Practice

1. Explain the main purpose of ipv6 fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 30: IPv6 Address Types

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

IPv6 defines several address scopes and purposes. Global unicast addresses are generally routable, link-local addresses work only on the local link, multicast addresses reach groups, and loopback ::1 refers to the local host.

Key points

- Every IPv6 interface commonly has a link-local address.
- Multicast replaces many broadcast functions.
- Unique local addresses can serve private-style internal designs.
- Prefix lengths are usually expressed in CIDR form.

Key terms and examples

Term	Meaning	Example
Global unicast	routable address	2001:db8::/32 docs
Link-local	one link only	fe80::1
Loopback	local host	::1

Practice

1. Explain the main purpose of ipv6 address types.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 31: IPv6 Neighbor Discovery and SLAAC

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

Neighbor Discovery Protocol performs local discovery functions in IPv6, including router discovery and address resolution. SLAAC lets hosts form addresses automatically using router advertisements.

Key points

- Router Advertisements describe prefixes and gateway information.
- Neighbor Solicitation/Advertisement replace ARP-style resolution.
- Duplicate Address Detection checks uniqueness.
- DHCPv6 may provide additional addressing or configuration.

Key terms and examples

Term	Meaning	Example
RA	router advertisement	prefix info
NDP	neighbor discovery	address resolution
SLAAC	automatic addressing	host self-config

Practice

1. Explain the main purpose of ipv6 neighbor discovery and slaac.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 32: NAT and PAT

Complete Networking Coursebook - explanations in English

IP ADDRESSING

Core idea

Network Address Translation modifies IP addressing information as traffic crosses a device. PAT, often called NAT overload, allows many private IPv4 hosts to share one public address by translating transport ports.

Key points

- NAT conserves public IPv4 addresses but changes end-to-end addressing.
- PAT tracks sessions using translated port numbers.
- NAT is not a substitute for a security policy.
- Some applications require special handling across translation.

Key terms and examples

Term	Meaning	Example
Static NAT	fixed mapping	server publication
Dynamic NAT	pool mapping	temporary public IP
PAT	many-to-one	home internet

Practice

1. Explain the main purpose of nat and pat.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Core idea

Routers forward packets between IP networks using routing tables. A route identifies a destination prefix and next-hop information or outgoing interface. The most specific matching route is normally chosen.

Key points

- Connected routes appear for active local interfaces.
- Static routes are manually configured.
- Dynamic protocols learn routes from other routers.
- A default route is used when no more-specific route exists.

Key terms and examples

Term	Meaning	Example
Connected	directly attached	LAN interface
Static	manual route	ip route ...
Default	catch-all route	0.0.0.0/0

Practice

1. Explain the main purpose of routing fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 34: Longest Prefix Match

Complete Networking Coursebook - explanations in English

ROUTING

Core idea

When multiple routes match a destination, routers prefer the route with the longest prefix, because it is the most specific network. Administrative and metric decisions matter when routes have comparable destination specificity.

Key points

- /24 is more specific than /16 for the same destination.
- Default routes have prefix length zero.
- Route summarization creates broader prefixes.
- Troubleshooting should inspect the actual forwarding table.

Key terms and examples

Term	Meaning	Example
/24	more specific	192.0.2.0/24
/16	less specific	192.0.0.0/16
/0	default	0.0.0.0/0

Practice

1. Explain the main purpose of longest prefix match.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 35: Static and Default Routes

Complete Networking Coursebook - explanations in English

ROUTING

Core idea

Static routes are simple and predictable but require manual maintenance. Default routes reduce table size for edge networks by forwarding unknown destinations toward an upstream router.

Key points

- Static routes suit small or stable networks.
- Floating static routes can serve as backups.
- Default routes are common at branch and internet edges.
- Incorrect next hops can create black holes or loops.

Key terms and examples

Term	Meaning	Example
Static route	manual path	branch LAN
Floating static	backup route	higher distance
Default route	upstream path	internet gateway

Practice

1. Explain the main purpose of static and default routes.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 36: Dynamic Routing Concepts

Complete Networking Coursebook - explanations in English

ROUTING

Core idea

Dynamic routing protocols exchange reachability information and calculate paths automatically. They differ in algorithm, convergence behavior, scalability, metrics and administrative scope.

Key points

- Interior protocols operate within an administrative domain.
- Exterior routing connects autonomous systems.
- Convergence is the time needed to reach consistent routing state.
- Metrics may consider hop count, cost, bandwidth or policy.

Key terms and examples

Term	Meaning	Example
IGP	inside domain	OSPF
EGP	between domains	BGP
Metric	path preference	OSPF cost

Practice

1. Explain the main purpose of dynamic routing concepts.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Core idea

OSPF is a link-state interior routing protocol. Routers build a topology database and calculate shortest paths using SPF logic. Areas help scale large OSPF deployments.

Key points

- OSPF exchanges link-state information rather than full tables periodically.
- Router IDs identify OSPF routers.
- Area 0 is the backbone area in multi-area designs.
- Cost is the main path metric.

Key terms and examples

Term	Meaning	Example
LSA	link-state information	topology update
Area	scaling boundary	area 0
SPF	shortest-path calculation	best route

Practice

1. Explain the main purpose of ospf basics.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 38: BGP Basics

Complete Networking Coursebook - explanations in English

ROUTING

Core idea

BGP is the main interdomain routing protocol of the internet. It exchanges network reachability between autonomous systems and applies routing policy using path attributes.

Key points

- BGP is policy-oriented rather than simply choosing the shortest physical path.
- AS_PATH records autonomous systems traversed.
- Prefixes may be filtered or preferred based on policy.
- BGP design requires careful security and operational controls.

Key terms and examples

Term	Meaning	Example
ASN	autonomous system number	network identity
AS_PATH	path attribute	AS sequence
Policy	route decision	prefer provider A

Practice

1. Explain the main purpose of bgp basics.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 39: Routing Troubleshooting

Complete Networking Coursebook - explanations in English

ROUTING

Core idea

Routing problems may come from missing routes, incorrect masks, bad next hops, protocol adjacency failures, filtering or asymmetric paths. Troubleshooting should compare intended topology with actual forwarding information.

Key points

- Check interface state before routing protocols.
- Verify addresses and subnet masks.
- Use ping and traceroute to locate failure points.
- Inspect route tables and protocol neighbors.

Key terms and examples

Term	Meaning	Example
No route	destination missing	packet dropped
Bad mask	wrong network size	misrouting
Adjacency	routing-peer state	OSPF neighbor

Practice

1. Explain the main purpose of routing troubleshooting.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 40: DHCP

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

DHCP automatically provides hosts with IP configuration such as address, prefix, gateway and DNS servers. In IPv4, a common sequence is Discover, Offer, Request and Acknowledge.

Key points

- DHCP reduces manual addressing errors.
- Leases allow addresses to be reused.
- Relay agents forward client requests across routers.
- Reservations can assign predictable addresses to known devices.

Key terms and examples

Term	Meaning	Example
DORA	DHCP exchange	Discover-Offer-Request-ACK
Lease	temporary assignment	8 hours
Relay	cross-subnet forwarding	helper address

Practice

1. Explain the main purpose of dhcp.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 41: DNS

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

The Domain Name System is a distributed hierarchical database that maps names to records such as IP addresses. Recursive resolvers query authoritative servers and cache answers for a defined TTL.

Key points

- A records map names to IPv4 addresses.
- AAAA records map names to IPv6 addresses.
- CNAME provides an alias.
- MX records identify mail servers.

Key terms and examples

Term	Meaning	Example
A	IPv4 mapping	host -> 192.0.2.1
AAAA	IPv6 mapping	host -> 2001:db8::1
MX	mail exchanger	domain mail server

Practice

1. Explain the main purpose of dns.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 42: NTP and Time Synchronization

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

Accurate time is critical for logs, authentication, certificates and troubleshooting. NTP synchronizes device clocks using hierarchical time sources.

Key points

- Consistent timestamps make event correlation possible.
- Large clock errors can break security protocols.
- Organizations often use internal NTP servers.
- Time sources should be redundant and monitored.

Key terms and examples

Term	Meaning	Example
NTP	time protocol	UDP 123
Stratum	source distance	stratum 1
Clock drift	time error growth	unsynchronized device

Practice

1. Explain the main purpose of ntp and time synchronization.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 43: HTTP, HTTPS and TLS

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

HTTP transfers web requests and responses. HTTPS protects HTTP using TLS, which provides encryption, integrity and server authentication using digital certificates.

Key points

- TLS certificates bind public keys to identities.
- HTTPS normally uses TCP port 443.
- Modern security requires current TLS versions and cipher choices.
- Certificate validation checks trust, names and validity periods.

Key terms and examples

Term	Meaning	Example
HTTP	web protocol	port 80
HTTPS	HTTP over TLS	port 443
Certificate	identity + public key	server cert

Practice

1. Explain the main purpose of http, https and tls.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 44: Email Protocols

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

Email systems use SMTP to transfer outgoing mail and protocols such as IMAP to access messages. Security features and authentication protect mail transport and user access.

Key points

- SMTP transfers messages between mail systems.
- IMAP keeps mail synchronized on a server.
- TLS can protect mail sessions.
- DNS records help route and authenticate mail.

Key terms and examples

Term	Meaning	Example
SMTP	mail transfer	server-to-server
IMAP	mail access	sync mailbox
MX	mail routing record	DNS

Practice

1. Explain the main purpose of email protocols.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 45: SSH, File Transfer and Remote Access

Complete Networking Coursebook - explanations in English

NETWORK SERVICES

Core idea

Administrators commonly use SSH for encrypted command-line access. Secure file-transfer methods include SFTP and SCP. Remote management should use strong authentication, restricted access and logging.

Key points

- SSH encrypts credentials and session content.
- Keys can be safer than reusable passwords when managed correctly.
- Management access should be limited to trusted networks or VPNs.
- Legacy plaintext protocols should be avoided.

Key terms and examples

Term	Meaning	Example
SSH	secure shell	port 22
SFTP	file transfer over SSH	secure upload
Key auth	public-key login	administrator access

Practice

1. Explain the main purpose of ssh, file transfer and remote access.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 46: Wi-Fi Fundamentals

Complete Networking Coursebook - explanations in English

WIRELESS

Core idea

Wi-Fi uses IEEE 802.11 standards to provide wireless LAN access. Performance depends on frequency band, channel width, modulation, signal strength, interference, client capability and access-point design.

Key points

- 2.4 GHz travels farther but has fewer non-overlapping channels.
- 5 GHz and 6 GHz provide more spectrum where supported.
- Actual throughput is lower than theoretical PHY rate.
- Wireless is a shared medium.

Key terms and examples

Term	Meaning	Example
SSID	network name	CampusWiFi
AP	access point	wireless bridge
Channel	radio frequency allocation	channel 36

Practice

1. Explain the main purpose of wi-fi fundamentals.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 47: RF Concepts

Complete Networking Coursebook - explanations in English

WIRELESS

Core idea

Wireless design requires understanding radio propagation. Signal strength decreases with distance and obstacles, while interference and noise reduce usable signal quality.

Key points

- RSSI indicates received signal strength.
- SNR compares signal with background noise.
- Walls and materials attenuate radio energy.
- Higher frequencies generally have shorter range through obstacles.

Key terms and examples

Term	Meaning	Example
RSSI	signal level	-55 dBm
SNR	signal-to-noise ratio	30 dB
Attenuation	signal loss	concrete wall

Practice

1. Explain the main purpose of rf concepts.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 48: Wi-Fi Security

Complete Networking Coursebook - explanations in English

WIRELESS

Core idea

Modern Wi-Fi security uses WPA2 or WPA3 with strong authentication and encryption. Enterprise deployments often integrate 802.1X authentication with a RADIUS service.

Key points

- Avoid obsolete WEP and weak shared passwords.
- WPA3 improves protections where devices support it.
- Guest traffic should be segmented from internal systems.
- Management frames and rogue access points should be monitored.

Key terms and examples

Term	Meaning	Example
WPA2	wireless security	AES/CCMP
WPA3	newer security	SAE
802.1X	enterprise authentication	RADIUS-backed

Practice

1. Explain the main purpose of wi-fi security.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 49: Wireless Roaming and Design

Complete Networking Coursebook - explanations in English

WIRELESS

Core idea

Enterprise WLANs use multiple access points to provide coverage and capacity. Roaming allows clients to move between APs while remaining connected, although the client usually decides when to roam.

Key points

- AP placement should consider both coverage and capacity.
- Excess transmit power can create sticky-client problems.
- Channel planning reduces co-channel interference.
- Site surveys validate design assumptions.

Key terms and examples

Term	Meaning	Example
Roaming	move between APs	office floor
Site survey	measure RF	heatmap
Capacity	clients per area	lecture hall

Practice

1. Explain the main purpose of wireless roaming and design.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 50: Wireless Troubleshooting

Complete Networking Coursebook - explanations in English

WIRELESS

Core idea

Wireless faults can be caused by weak signal, interference, channel congestion, authentication failures, driver issues or upstream network problems. Troubleshooting should separate RF problems from IP and service problems.

Key points

- Check whether the client associates with the intended AP.
- Measure RSSI and SNR.
- Verify DHCP, gateway and DNS after association.
- Compare affected and unaffected clients.

Key terms and examples

Term	Meaning	Example
Association	join AP	802.11 connection
Interference	competing RF	neighbor AP
Captive portal	web authentication	guest Wi-Fi

Practice

1. Explain the main purpose of wireless troubleshooting.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 51: Network Security Principles

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

Network security protects confidentiality, integrity and availability. Effective defense uses layered controls, least privilege, segmentation, secure configuration, monitoring, backups and incident response.

Key points

- No single control eliminates all risk.
- Segmentation limits lateral movement.
- Patching reduces exposure to known vulnerabilities.
- Logs and alerts support detection and investigation.

Key terms and examples

Term	Meaning	Example
CIA	security objectives	confidentiality/integrity/availability
Least privilege	minimum access	restricted admin rights
Defense in depth	multiple controls	firewall + MFA

Practice

1. Explain the main purpose of network security principles.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 52: Firewalls and ACLs

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

Firewalls and access-control lists permit or deny traffic according to policy. Stateful firewalls track connection state, while simple ACLs often filter using addresses, protocols and ports.

Key points

- Rules should follow documented business needs.
- Specific rules should be reviewed before broad permit rules.
- Implicit deny behavior should be understood.
- Logging helps validate and troubleshoot policy.

Key terms and examples

Term	Meaning	Example
ACL	packet filter	permit TCP 443
Stateful	tracks sessions	return traffic allowed
Rule order	processing sequence	top-down

Practice

1. Explain the main purpose of firewalls and acls.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 53: VPNs

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

Virtual private networks create protected tunnels across untrusted networks. Site-to-site VPNs connect networks, while remote-access VPNs connect individual users.

Key points

- Encryption protects data in transit.
- Authentication verifies peers or users.
- Split tunneling changes which traffic uses the VPN.
- Key management and endpoint security remain important.

Key terms and examples

Term	Meaning	Example
IPsec	network-layer VPN	site-to-site
TLS VPN	secure remote access	user portal
Tunnel	encapsulated path	encrypted connection

Practice

1. Explain the main purpose of vpns.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 54: AAA, RADIUS and TACACS+

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

AAA stands for authentication, authorization and accounting. Centralized AAA systems improve access control and auditing for network users and administrators.

Key points

- Authentication verifies identity.
- Authorization determines permitted actions.
- Accounting records activity.
- RADIUS is widely used for network access; TACACS+ is common in device administration.

Key terms and examples

Term	Meaning	Example
Authentication	who are you?	password + MFA
Authorization	what may you do?	admin role
Accounting	what happened?	login log

Practice

1. Explain the main purpose of aaa, radius and tacacs+.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 55: Common Network Attacks

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

Network attacks include spoofing, scanning, credential theft, denial of service, man-in-the-middle techniques, malicious DNS manipulation and exploitation of vulnerable services.

Key points

- Attack prevention begins with reducing unnecessary exposure.
- Strong authentication reduces credential abuse.
- Segmentation and filtering reduce attack paths.
- Monitoring should focus on abnormal behavior and known indicators.

Key terms and examples

Term	Meaning	Example
Spoofing	fake identity/address	forged source
DoS	resource exhaustion	traffic flood
MITM	intercept communication	rogue gateway

Practice

1. Explain the main purpose of common network attacks.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 56: IDS, IPS and Network Monitoring

Complete Networking Coursebook - explanations in English

SECURITY

Core idea

Intrusion detection systems analyze activity for signs of attacks, while intrusion prevention systems can block selected traffic. Effective monitoring combines logs, flows, packet captures, metrics and alerts.

Key points

- Signature detection finds known patterns.
- Behavior analysis looks for unusual activity.
- False positives require tuning.
- Monitoring data should be time-synchronized and protected.

Key terms and examples

Term	Meaning	Example
IDS	detects suspicious activity	alert
IPS	blocks selected activity	inline prevention
Flow log	connection metadata	NetFlow-style record

Practice

1. Explain the main purpose of ids, ips and network monitoring.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 57: Cloud Networking

Complete Networking Coursebook - explanations in English

OPERATIONS & CLOUD

Core idea

Cloud platforms provide virtual networks, subnets, routing, load balancers, security groups, gateways and private connectivity. Concepts resemble traditional networking but are implemented through software-defined controls.

Key points

- Cloud subnets are logical address ranges.
- Security groups filter traffic at virtual interfaces.
- Load balancers distribute application traffic.
- Hybrid networks connect cloud and on-premises environments.

Key terms and examples

Term	Meaning	Example
VPC/VNet	virtual network	cloud network
Security group	virtual firewall	instance policy
Load balancer	traffic distribution	web tier

Practice

1. Explain the main purpose of cloud networking.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 58: Software-Defined Networking

Complete Networking Coursebook - explanations in English

OPERATIONS & CLOUD

Core idea

Software-defined networking separates control decisions from packet forwarding and uses programmable interfaces to automate network behavior. It supports centralized policy, automation and rapid change.

Key points

- Controllers maintain a broader view of network state.
- APIs enable orchestration and automation.
- Intent-based systems translate policy goals into configuration.
- Automation requires validation, testing and rollback plans.

Key terms and examples

Term	Meaning	Example
Controller	control logic	central platform
API	automation interface	REST API
Intent	desired state	segment guest users

Practice

1. Explain the main purpose of software-defined networking.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 59: Troubleshooting Methodology

Complete Networking Coursebook - explanations in English

OPERATIONS & CLOUD

Core idea

Good troubleshooting is systematic: define the problem, gather evidence, identify the likely layer, test a hypothesis, apply the smallest safe change, verify the result and document what was learned.

Key points

- Begin with scope: one host, one VLAN, one site or everyone?
- Compare working and failing paths.
- Change one variable at a time when possible.
- Preserve logs and evidence before disruptive changes.

Key terms and examples

Term	Meaning	Example
Scope	who/what affected	one subnet
Hypothesis	possible cause	DNS failure
Verification	confirm fix	repeat test

Practice

1. Explain the main purpose of troubleshooting methodology.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Lesson 60: Network Design and Final Review

Complete Networking Coursebook - explanations in English

OPERATIONS & CLOUD

Core idea

Network design combines requirements, topology, addressing, redundancy, security, capacity, operations and cost. A strong design is understandable, scalable, documented and testable.

Key points

- Start from business and application requirements.
- Design for failure, not only normal operation.
- Use hierarchical and modular patterns where useful.
- Document assumptions, dependencies and acceptance tests.

Key terms and examples

Term	Meaning	Example
Requirement	design need	99.9% availability
Redundancy	backup path/device	dual uplinks
Acceptance test	proof of outcome	failover test

Practice

1. Explain the main purpose of network design and final review.
2. Give one practical networking example related to this lesson.
3. Identify one mistake or risk that a network administrator should avoid.
4. Write two technical terms from this lesson and define them in your own words.

Exam focus

- Connect this topic to the OSI/TCP-IP layer where it mainly operates.
- Explain how the concept affects performance, reliability or security.
- Use one precise example, address, protocol or device in your answer.

Study notes / diagram

Overview

Prefixes frequently used in LAN design.

Key reference

Item	Value / concept	Use / meaning
/24	255.255.255.0	256 addresses / 254 traditional usable
/25	255.255.255.128	128 / 126
/26	255.255.255.192	64 / 62
/27	255.255.255.224	32 / 30
/28	255.255.255.240	16 / 14
/29	255.255.255.248	8 / 6
/30	255.255.255.252	4 / 2

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Memorize the service purpose first; port numbers help interpret firewall rules and packet captures.

Key reference

Item	Value / concept	Use / meaning
DNS	53 TCP/UDP	Name resolution
DHCP	67/68 UDP	IPv4 address configuration
HTTP	80 TCP	Web
HTTPS	443 TCP	Secure web
SSH	22 TCP	Secure remote shell
SMTP	25 TCP	Mail transfer
NTP	123 UDP	Time synchronization

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Use layers as a diagnostic framework.

Key reference

Item	Value / concept	Use / meaning
Physical	cable, radio, link light	Is the interface up?
Data Link	VLAN, MAC, STP	Is local switching correct?
Network	IP, mask, routes	Can the destination network be reached?
Transport	ports, TCP/UDP	Is the service reachable?
Application	DNS/app/config	Does the service itself work?

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

A secure network combines architecture, configuration and operations.

Key reference

Item	Value / concept	Use / meaning
Access	MFA, least privilege	limit administration
Segmentation	VLANs/firewalls	reduce lateral movement
Updates	patching	close known vulnerabilities
Monitoring	logs/IDS	detect abnormal activity
Recovery	backups/config copies	restore after failure

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Wireless Design Checklist

Complete Networking Coursebook - explanations in English

REVISION

Overview

Plan coverage, capacity and security together.

Key reference

Item	Value / concept	Use / meaning
Coverage	RSSI/SNR targets	avoid dead zones
Capacity	client density	size AP count
Channels	reuse/interference	plan spectrum
Security	WPA2/WPA3	protect access
Validation	site survey	measure real environment

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

A user can reach the default gateway but cannot open websites by name. What should be checked next?

Key reference

Item	Value / concept	Use / meaning
Evidence	gateway reachable	local IP path works
Likely area	DNS	name resolution
Tests	nslookup / DNS ping	compare name vs IP
Action	verify DNS server	DHCP/static settings

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Troubleshooting Scenario 2

Complete Networking Coursebook - explanations in English

PRACTICE

Overview

Hosts in VLAN 10 can communicate with each other but not with VLAN 20.

Key reference

Item	Value / concept	Use / meaning
Layer 2	local switching works	same VLAN communication
Layer 3	inter-VLAN routing	required between VLANs
Checks	SVI/router interfaces	status and addresses
Policy	ACL/firewall	may block traffic

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Troubleshooting Scenario 3

Complete Networking Coursebook - explanations in English

PRACTICE

Overview

Wi-Fi users connect successfully but experience poor voice quality during busy periods.

Key reference

Item	Value / concept	Use / meaning
Symptoms	jitter/latency	real-time sensitivity
RF	channel utilization	congestion/interference
Capacity	AP/client load	too many clients
QoS	voice prioritization	traffic handling

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Revision Test A

Complete Networking Coursebook - explanations in English

ASSESSMENT

Overview

Answer without notes, then check against the review guide.

Key reference

Item	Value / concept	Use / meaning
Q1	Explain OSI vs TCP/IP	layering
Q2	Subnet 192.168.50.0/24 into /27s	addressing
Q3	Explain VLAN + trunk	switching
Q4	Compare TCP and UDP	transport
Q5	Describe DHCP and DNS roles	services

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Use diagrams and precise technical vocabulary.

Key reference

Item	Value / concept	Use / meaning
Q1	Explain longest prefix match	routing
Q2	Describe OSPF vs BGP	dynamic routing
Q3	Explain WPA3 and 802.1X	wireless security
Q4	Describe a site-to-site VPN	security
Q5	Create a troubleshooting workflow	operations

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Design a small office network with two VLANs, internet access and Wi-Fi.

Key reference

Item	Value / concept	Use / meaning
VLAN 10	Staff	192.168.10.0/24
VLAN 20	Guest	192.168.20.0/24
Gateway	router/L3 switch	inter-VLAN routing
DHCP	one pool per VLAN	automatic addressing
Security	guest isolation	deny access to staff LAN

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Allocate VLSM subnets for 100, 50, 20 and 2-host requirements from a /24 block.

Key reference

Item	Value / concept	Use / meaning
100 hosts	/25	126 usable
50 hosts	/26	62 usable
20 hosts	/27	30 usable
2 hosts	/30	2 usable

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Use a packet capture to identify Ethernet, IP, TCP/UDP and application information.

Key reference

Item	Value / concept	Use / meaning
Ethernet	source/destination MAC	Layer 2
IP	source/destination IP	Layer 3
TCP/UDP	ports	Layer 4
Application	DNS/HTTP/etc.	upper layer

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Selected networking terms.

Key reference

Item	Value / concept	Use / meaning
Broadcast domain	group receiving Layer 2 broadcasts	usually one VLAN
Default gateway	router used for remote networks	host setting
MTU	maximum frame/packet payload limit	path consideration
Prefix	network bits	/24
Throughput	achieved transfer rate	measured performance

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.

Overview

Use this guide to self-check the major ideas from the course.

Key reference

Item	Value / concept	Use / meaning
Layering	encapsulation and responsibilities	OSI/TCP-IP
Switching	MAC, VLAN, STP	LAN behavior
Addressing	IPv4/IPv6/subnetting	logical identity
Routing	best path / longest prefix	inter-network forwarding
Security	segmentation, filtering, monitoring	risk reduction

Study / answer space

Write the most important rule or principle from this page.

Add one diagram, calculation or example from memory.

State one troubleshooting or security implication.